Prasad Anand

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EDUCATION

Zeal College of Engineering and Research, Pune (SPPU) 2024 – 2028
B.Tech, Artificial Intelligence & Machine Learning *Pune, MH*
CGPA: 8.97/10 | Sem I: 8.45 | Sem II: 9.38 | Sem III: 9.32 | Sem IV: 8.64

Silver Crest Junior College 2022 – 2024
Higher Secondary Education (Class XII) | *Percentage: 71.5%* *Pune, MH*

Dnyanganga English Medium School 2009 – 2022
Secondary Education (Class X) | *Percentage: 85.8%* *Pune, MH*

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL, HTML/CSS
Backend & Web: FastAPI, React.js, Node.js, Vite, TailwindCSS
Databases: PostgreSQL, MySQL, Supabase
Core Concepts: Object-Oriented Programming (OOP), Database Normalization, REST API Design
ML / Data: Machine Learning, Deep Learning, PyTorch, Scikit-Learn, Hugging Face Transformers, pandas, NumPy, Matplotlib, Seaborn, LightGBM, Natural Language Processing (NLP), Optical Character Recognition (OCR)
Tools: Git, GitHub, CI/CD, Linux, Vercel, Ollama, Jupyter Notebook, OpenDrift, MATLAB

PROJECTS

OilTrace: SAR-Based Oil Spill Detection & Source Attribution
Python, PyTorch, U-Net, OpenDrift, pandas, AIS

- Built a **3-stage pipeline** for NTRO's **Smart India Hackathon 2026** problem statement: **SAR oil-slick detection**, **backward drift tracing**, and **AIS vessel ranking**, chained through importable modules and a single pipeline entry point.
- **Detection:** Trained a single-channel **U-Net (31.04M parameters)** on **3,354** Sentinel-1 SAR image/mask pairs and evaluated on **839** test pairs, achieving **IoU 0.6630**; trained **10 epochs** on an RTX 5060 laptop GPU and **90 epochs** on a rented RTX 5090 with per-epoch checkpointing.
- **Drift Trace:** Implemented a **physics-based Lagrangian simulation** in **OpenDrift** using real wind (OpenWeatherMap) and ocean current (Copernicus Marine) data to estimate the slick's **origin point and time window**, with forward spread prediction and coastline stranding handling.
- **Attribution:** Built AIS filtering and weighted scoring (proximity, trajectory alignment, speed drops, AIS gaps, vessel type) to produce an **explainable ranked suspect list**; the planted vessel ranked **#1 of 14** in a synthetic Persian Gulf scenario and across **25 randomized seeds**.
- Documented limitations: **false positives** on look-alike scenes, drift accuracy bounded by wind/current data quality, and attribution as a probabilistic ranking, not a definitive result.
- **GitHub:** github.com/prasadanand412/oiltrace

Blood Bank Management System
React.js, TypeScript, FastAPI, PostgreSQL, SQLAlchemy, JWT

- Built a **full-stack** blood bank platform with **role-based access control** (Admin, Staff, Hospital, Donor), a SaaS analytics dashboard, and priority-based request workflows (Emergency, High, Medium, Low).
- Designed an **11-table normalized PostgreSQL** database (3NF/BCNF) with **5 inventory triggers**, **3 FIFO allocation procedures**, and optimized aggregation views.
- Developed **12 REST endpoints** with **FastAPI**, **SQLAlchemy**, **JWT** authentication, and Pydantic validation; enforced 90-day donor eligibility.
- Built the **React + TypeScript** frontend with TailwindCSS, shadcn/ui, and Recharts, including KPI cards.
- **GitHub:** github.com/prasadanand412/blood-bank-management-system

DocuSense: AI-Powered Document Analysis System
Python, OCR, Qwen2.5:7B, Gemini API, NLP

- Built a **document intelligence pipeline** to extract, process, classify, and summarize information from PDFs and scanned documents.
- Integrated **OCR** for scanned documents and local **Qwen2.5:7B** inference via **Ollama** for private, offline processing.
- Used **Gemini API** for contextual summarization and designed a **multi-stage NLP workflow** for classification and key-entity extraction.
- **GitHub:** github.com/prasadanand412/DocuSense-document-extractor

Wellness+: Health Monitoring Dashboard
React.js, Vite, Supabase

- Designed and built the **user onboarding flow** and **interactive dashboard** for a team-built healthcare web app to monitor health metrics, manage wellness records, and visualize trends.
- Set up the **Supabase** database and contributed to **backend** development, including user authentication and secure storage of health data.
- **GitHub:** github.com/prasadanand412/wellness

EXPERIENCE

Cubeage Technologies Pvt Ltd. Jun 2026 – Aug 2026
ML Intern *Pune, MH*

- Designed and developed a **document analysis and classification system** using **BART-Large-MNLI** from Hugging Face for automated document categorization.
- Applied **NLP techniques** including **tokenization**, **n-grams**, **text preprocessing**, and **feature extraction** to real-world document processing tasks; studied **Retrieval-Augmented Generation (RAG)** and **LLM**-based response generation.
- Worked with **Hugging Face Transformers** and pretrained models to build and experiment with **text classification** pipelines.

Tantrata Solutions Pvt. Ltd. Jan 2026 – Feb 2026
Software Development Intern *Pune, MH*

- Built and deployed a responsive personal portfolio website using **React.js**, **Vite**, HTML, CSS, and **JavaScript** (live: prasadanand.in).
- Implemented **Progressive Web App (PWA)** features including service workers and manifest configuration for offline support and cross-device access.
- Configured **CI/CD**-style deployment pipelines using **GitHub** and **Vercel**, enabling automated builds and zero-downtime production deployments.

Greenfinch Digital Business Solutions Jun 2025 – Jul 2025
Business Development Intern *Pune, MH*

- Performed **data-driven market research** and prospective client analysis to identify high-value business opportunities across multiple industry segments.
- Built and maintained structured **lead-tracking spreadsheets** and prepared **business reports** tracking outreach performance metrics.
- Coordinated client outreach and cross-functional communication, supporting a more efficient **client acquisition pipeline**.

ACHIEVEMENTS

- **National Science Olympiad** — **Silver Medalist** in school.
- Nominated by school faculty for **full-fee sponsorship** to compete in the **State-Level Homi Bhabha Science Examination**.
- Completed multiple certifications on **Kaggle** and **HackerRank**; completed the **Deloitte Data Analytics Job Simulation**.